# Physical Quantities and Measurement Techniques

📂 1.1 Physical Quantities and Measurement Techniques 🎯 IGCSE Physics 0625 — Extended 📝 Version 1 📅 2026-06-30

## 📜📜 History

Measurement is fundamental to physics — without measurement, physics is just philosophy.

**Ancient measurements** were based on human body parts: the cubit (forearm), the foot, the hand. These varied from person to person, causing confusion in trade and construction.

**The Metric System** was born during the French Revolution (1790s). Scientists defined the **metre** as one ten-millionth of the distance from the North Pole to the Equator. This was the first standardised system based on natural constants.

**Modern definitions** are even more precise:
- The **metre** is now defined by the speed of light in a vacuum ( m/s)
- The **second** is defined by the vibration of caesium-133 atoms ( oscillations)
- The **kilogram** was redefined in 2019 using Planck's constant ()

**Did you know?** The original "kilogram" prototype — a cylinder of platinum-iridium alloy kept in France — was the last physical artefact used to define an SI unit until 2019. It was affectionately called "Le Grand K".

$c = 299\,792\,458$ $9\,192\,631\,770$ $h$

## 🌍 🌍 Nature & Applications

Measurement is everywhere:

- 📏 **Length**: Measuring height, distance, width of a hair ( mm)
- ⏱️ **Time**: Stopwatches for races, atomic clocks for GPS satellites
- ⚖️ **Mass**: Kitchen scales, laboratory balances, weighing trucks
- 🌡️ **Temperature**: Thermometers in medicine, weather, cooking
- 📊 **Volume**: Measuring cylinders for liquids, displacement method for irregular solids

**Why standard units matter**: If a doctor says "take 5 mL of medicine" or an engineer says "this bolt is 10 mm", everyone in the world knows exactly what that means.

$\approx 0.1$

## ⚛️ ⚛️ Theory

### Base SI Quantities

| Base Quantity | SI Unit | Symbol |
|--------------|---------|--------|
| Length | metre | m |
| Mass | kilogram | kg |
| Time | second | s |
| Temperature | kelvin | K |
| Electric current | ampere | A |
| Amount of substance | mole | mol |
| Luminous intensity | candela | cd |

For IGCSE Physics, the most important are: **length (m), mass (kg), time (s), and temperature (°C or K)**.

### Prefixes

| Prefix | Symbol | Factor | Example |
|--------|--------|--------|---------|
| kilo- | k | | 1 km = 1000 m |
| centi- | c | | 1 cm = 0.01 m |
| milli- | m | | 1 mm = 0.001 m |
| micro- | | | 1 m = 0.000 001 m |
| nano- | n | | 1 nm = 0.000 000 001 m |

### Measuring Instruments

| Instrument | Measures | Precision | How |
|-----------|----------|-----------|-----|
| Metre ruler | Length | mm | Read at eye level to avoid parallax |
| Measuring tape | Length | mm | For curves / long distances |
| Vernier callipers | Small lengths | mm | Internal/external/depth measurements |
| Micrometer screw gauge | Very small lengths | mm | Thickness of wire/sheet |
| Measuring cylinder | Volume | mL | Read bottom of meniscus |
| Stopwatch | Time | s | Digital or analogue |
| Top-pan balance | Mass | g | Zero before use |
| Thermometer | Temperature | °C | Bulb fully immersed |

### Measuring Volume of Irregular Solids

Use the **displacement method**:

1. Fill a measuring cylinder with a known volume of water
2. Submerge the object completely
3. Read the new water level
4. Subtract:

### Accuracy vs Precision

- **Accuracy**: How close a measurement is to the true value
- **Precision**: How consistent repeated measurements are

A measurement can be precise but inaccurate (e.g., a ruler that's slightly stretched).

Always take **repeat readings** and calculate the average to reduce random errors.

### Uncertainty

For a ruler with mm markings, the uncertainty is mm.

$$\text{Value} = \text{number} \times 10^{\text{power}}$$

$$V_{\text{object}} = V_{\text{final}} - V_{\text{initial}}$$

$$\text{Average} = \frac{\text{sum of measurements}}{\text{number of measurements}}$$

$$\text{Uncertainty} = \frac{\text{range}}{2} = \frac{\text{max} - \text{min}}{2}$$

$10^3$ $10^{-2}$ $10^{-3}$ $\mu$ $10^{-6}$ $\mu$ $10^{-9}$ $\pm 1$ $\pm 1$ $\pm 0.1$ $\pm 0.01$ $\pm 0.5$ $\pm 0.1$ $\pm 0.1$ $\pm 0.5$ $V_{\text{object}} = V_{\text{final}} - V_{\text{initial}}$ $\pm 1$ $\pm 0.5$

## ✍️ ✍️ Worked Examples

### Example 1: Unit Conversion

Convert 2500 mm into metres.

**Solution:**

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### Example 2: Measuring Volume by Displacement

A stone is placed in a measuring cylinder containing 40 mL of water. The water level rises to 55 mL. What is the volume of the stone?

**Solution:**

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### Example 3: Calculating Uncertainty

A student measures the length of a table five times: 152 cm, 151 cm, 153 cm, 152 cm, 152 cm.

**(a)** What is the average length?
**(b)** What is the uncertainty?

**Solution (a):**

**Solution (b):**

Result: cm

$$1 \text{ m} = 1000 \text{ mm}$$

$$2500 \text{ mm} = \frac{2500}{1000} = 2.5 \text{ m}$$

$$V_{\text{stone}} = 55 - 40 = 15 \text{ cm}^3$$

$$\text{Average} = \frac{152 + 151 + 153 + 152 + 152}{5} = \frac{760}{5} = 152 \text{ cm}$$

$$\text{Uncertainty} = \frac{153 - 151}{2} = \pm 1 \text{ cm}$$

$152 \pm 1$